



BLUE TEAM SCENARIO

Threat Group	APT28 (Adversary Emulation/Fancy Bear/Sofacy)
Attribution	Russian GRU (Unit 26165)
Tactics	Credential Harvesting + Exfiltration
Target Industry	Finance / Government / Energy

Report ID :19-acquiring memory(47_48_49)-17_golden

Report : 2

Target :iman99.local (Windows 10 x64)

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Severity Summary: Critical: 1 | High: 1 | Medium: 0 | Low: 0

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Introduction

This paper documents a thorough P intended to assess the security infrastructure's detection and response capabilities and model a targeted cyberattack. With an emphasis on strategies used by advanced threat groups like APT28 (Operation BlueBalance), the scenario mimics an Advanced Persistent Threat (APT) lifecycle and maps directly to the MITRE ATT&CK architecture.

Validating end-to-end visibility across various attack vectors—from initial access via email phishing to endpoint compromise and subsequent credential theft memory manipulation—is the aim of this simulated experiment.

Scenario Description

The Attack

On a routine Tuesday morning, a senior analyst at the National Finance Authority receives an urgent email from "IT Security." The email contains an attachment named NulltackKatz.py and threatens system lockout if not installed immediately. The analyst, concerned about salary processing, downloads and runs the attachment. Unknown to them, the tool performs:

- Discovery — Maps local users, network configuration, and running processes
- Privilege Escalation — Enables SeDebugPrivilege for LSASS access
- Credential Dumping — Executes Mimikatz to extract passwords and NTLM hashes
- Exfiltration — Sends stolen credentials to an external Gmail account

Required Tools

Tool	Purpose	Where to Get
Sysmon	Endpoint monitoring	Microsoft Sysinternals
Wireshark	Network traffic capture	wireshark.org
Dumplt	Memory acquisition	Magnet Forensics
Volatility 3	Memory forensics	https://github.com/AnmaR3B/volatility3
Mimikatz	Credential dumping (simulated)	https://github.com/AnmaR3B/mimikatz
NulltackKatz.py	APT28 emulation tool	https://drive.google.com/file/d/1tpFjOmPWBGw21YLneFRgqN6st8lehZd5/view?usp=drive_link

Part A: Setup

Step 1 — Create Evidence Directory

```
(c) Microsoft Corporation. All rights reserved.  
C:\Users\iman SOC>mkdir C:\NulltackLab  
C:\Users\iman SOC>mkdir C:\NulltackLab\logs  
C:\Users\iman SOC>mkdir C:\NulltackLab\evidence
```

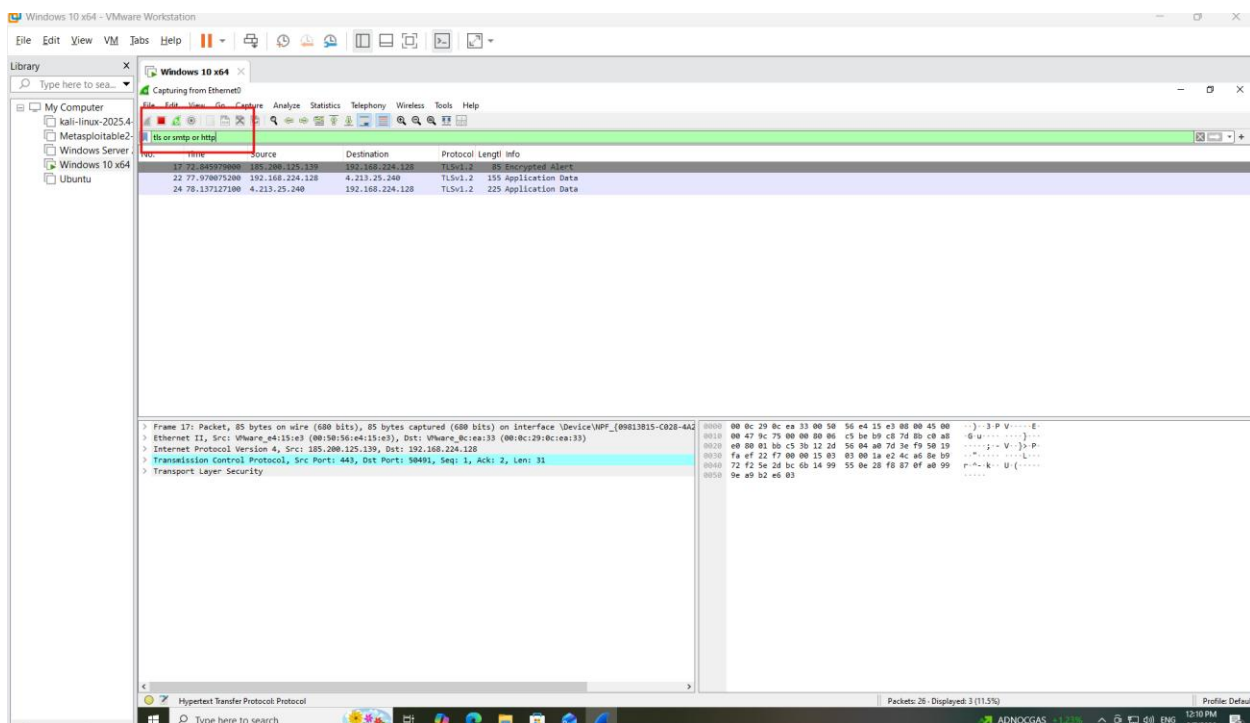
Step 2 — Download NulltackKatz.py

Save the tool as C:\NulltackLab\NulltackKatz.py

Step 3 - Configure Gmail for Exfiltration

Step 4 — Start Monitoring Stack

- Start Wireshark capture on all interfaces
- Filter: tls or smtp or http



- Verify Sysmon is running
- Get-Service Sysmon

```
PS C:\Users\iman SOC\Downloads> cd "C:\Users\iman SOC\Downloads\Sysmon"
PS C:\Users\iman SOC\Downloads\Sysmon> Get-Service Sysmon64

Status   Name      DisplayName
-----
Running  Sysmon64  Sysmon64
```

Part B: Execute Attack

Task 1 — Run NulltackKatz

```
Administrator: Windows PowerShell

[-] Log directory created: C:\NulltackLab\logs

=====
NULLTACK - APT28 EMULATION
=====
APT28 (BlueDelta / Fancy Bear)
Russian GRU - Unit 26165
T1566.001 --> T1003.001 --> T1041
Spearphishing LSASS Dump Exfiltration
=====

[API] APT Group: APT28 (BlueDelta / Fancy Bear / Sednit)
[API] Attribution: Russian GRU (Unit 26165)
[API] Campaign ID: BLUEDELTA_2026007_123942
[API] Target: FINANCE DEPARTMENT
[EMAIL] Exfiltration Email: pr.i.c.e0.025@gmail.com
=====
LAB INSTRUCTIONS FOR STUDENTS:
=====
Before running this tool, ensure your monitoring stack is active:
[OK] Wireshark capturing traffic (filter: tls or smtp)
[OK] Sysmon running (check: Get-Service Sysmon)
[OK] Process Monitor ready (optional)
[OK] DumpIt ready for memory capture
After simulation completes:
-> Check Email for exfiltrated credentials
-> Analyze Sysmon Event ID 10 (LSASS access)
-> Review the exfiltration report in logs
Press ENTER to begin APT28 simulation...

=====
[PHASE 1] T1566.001 - Spearphishing Attachment (APT28 Style)
=====
PHISHING EMAIL CONTENT
=====
From: IT Security <amearg@nulltack.com>
To: Finance Department <finance@iman999.local>
Subject: URGENT: Payroll System Security Update Required

Dear Finance Team,

Microsoft has released a critical security update for the Payroll
Verification System. Failure to install this update by EOD will
result in system lockout and delayed salary processing.

WARNING: Download and run the attached "NulltackKatz"
immediately to verify your credentials.

Thank you,
IT Security Team
```

Part C: Detection & Investigation

Analyze the generated phishing email:

- Open: C:\NulltackLab\logs\phishing_email_BLUEDELTA_*.txt

150611_114629.txt 6/11/2026 11:42 AM Text Document 1 KB

phishing_email_BLUEDELTA_20260611_114629.txt - Notepad

File Edit Format View Help

=====

PHISHING EMAIL CONTENT

=====

From: IT Security <security@payroll-update.com>
 Reply-To: it-security@noreply.com
 To: Finance Department <finance@iman999.local>
 Subject: URGENT: Payroll System Security Update Required

Dear Finance Team,

Microsoft has released a critical security update for the Payroll Verification System. Failure to install this update by EOD will result in system lockout and delayed salary processing.

WARNING: Download and run the attached "NulltackKatz.py" immediately to verify your credentials.

Thank you,
 IT Security Team

[Attachment: NulltackKatz.py]

=====

Task 2 - Email Analysis (T1566.001 Detection)

Analyze the generated phishing email:

- Open: C:\NulltackLab\logs\phishing_email_BLUEDELTA_*.txt

Question	Answer
What is the sender's display name?	IT Security
What is the actual reply-to address?	it-security@noreply.com
What urgent claim is used?	Failure to install this update by EOD will result in system lockout and delayed salary processing.
What MITRE technique ID covers this attack vector?	T1566.001 –(Spearphishing Attachment.)

Task 3 — Sysmon Analysis (T1003.001 Detection)

- Query the Sysmon Event Log for LSASS access:
- `Get-WinEvent -FilterHashtable @{'LogName='Microsoft-Windows-Sysmon/Operational'; ID=10 } | Where-Object {$_.Message -like "*lsass.exe*"} | Format-List`

```
[Press Enter to exit...]
PS C:\NulltackLab> Get-WinEvent -FilterHashtable @{'LogName='Microsoft-Windows-Sysmon/Operational'; ID=10 } | Where-Object {$_.Message -like "*lsass.exe*"} | Format-List
```

```
TargetUser: NT AUTHORITY\SYSTEM
TimeCreated : 6/9/2025 11:38:55 AM
ProviderName : Microsoft-Windows-Sysmon
Id : 10
Message : Process accessed:
RuleName :
UtcTime: 2026-06-09 18:38:55.764
SourceProcessGUID: {9fed14d4-4b38-6a28-6c00-000000001a00}
SourceProcessId: 4692
SourceThreadId: 18656
SourceImage: C:\Windows\system32\wbem\wmiprvse.exe
TargetProcessGUID: {9fed14d4-4b38-6a28-6c00-000000001a00}
TargetProcessId: 672
TargetImage: C:\Windows\system32\lsass.exe
GrantedAccess: 0x1410
CallTrace: C:\Windows\SYSTEM32\ntdll.dll+9d6d[C:\Windows\System32\KERNELBASE.dll+2829[C:\Windows\system32\wbem\cmd.exe+32.dll+fc40[C:\Windows\SYSTEM32\framedynos.dll+5d2[C:\Windows\SYSTEM32\framedynos.dll+7429[C:\Windows\system32\wbem\wmiprvse.exe+496[C:\Windows\system32\wbem\wmiprvse.exe+8c3[C:\Windows\System32\RPCRT4.dll+77a0[C:\Windows\System32\RPCRT4.dll+12567[C:\Windows\System32\combase.dll+bc66[C:\Windows\System32\RPCRT4.dll+56cb[C:\Windows\System32\combase.dll+983b[C:\Windows\System32\combase.dll+9813[C:\Windows\System32\combase.dll+bf36[C:\Windows\System32\combase.dll+52b7[C:\Windows\System32\combase.dll+acbf[C:\Windows\System32\combase.dll+3046[C:\Windows\System32\combase.dll+fd30[C:\Windows\System32\combase.dll+3d49[C:\Windows\System32\RPCRT4.dll+5678[C:\Windows\System32\RPCRT4.dll+3510[C:\Windows\System32\RPCRT4.dll+3d30[C:\Windows\System32\RPCRT4.dll+41ac
SourceUser: NT AUTHORITY\NETWORK SERVICE
TargetUser: NT AUTHORITY\SYSTEM
```

Question	Answer
What is the SourceImage that accessed lsass.exe?	C:\Windows\system32\wbem\wmiprvse.exe
What is the GrantedAccess value?	0x1410
What does 0x1010 represent?	The three permissions—PROCESS_VM_READ (0x0010), PROCESS_QUERY_INFORMATION (0x0400), and PROCESS_QUERY_LIMITED_INFORMATION (0x1000)—are combined to indicate the shared critical access that permits access to read the virtual memory of processes and query their data.
What is the TargetImage?	C:\Windows\system32\lsass.exe
Why is this event the PRIMARY IoC for T1003.001?	Because processes that request permissions to read memory (PROCESS_VM_READ) with a value equal to 0x1410 or 0x1010 on a sensitive system process such as (lsass.exe) is the IOC evidence on (Credential Dumping)

Task 4 — Process Tree Analysis

Query process creation events:

```
PS C:\NulltackLab> Get-WinEvent -FilterHashtable @{'LogName='Microsoft-Windows-Sysmon/Operational'; ID=1 } | Where-Object {
>> } | Where-Object {
>> $_.Message -like "*mimikatz*" -or $_.Message -like "*NulltackKatz*"
>> } | Format-List
```



```

ParentCommandLine: "C:\Users\Iman SOC\AppData\Local\Programs\Python\Launcher\py.exe" .\NulltackKatz_v1.2.py
ParentUser: IMAN999\Iman SOC
TimeCreated : 6/8/2026 11:45:39 PM
ProviderName : Microsoft-Windows-Sysmon
Id : 1
Message : Process Create:
RuleName:
UtcTime: 2026-06-09 06:45:39.606
ProcessGuid: {9fed14d4-b693-6a27-7902-000000001900}
ProcessId: 7696
Image: C:\Users\Iman SOC\AppData\Local\Programs\Python\Launcher\py.exe
FileVersion: 3.14.5
Description: Python
Product: Python
Company: Python Software Foundation
OriginalFileName: py.exe
CommandLine: "C:\Users\Iman SOC\AppData\Local\Programs\Python\Launcher\py.exe" .\NulltackKatz_v1.2.py
CurrentDirectory: C:\NulltackLab\
User: IMAN999\Iman SOC
LogonGuid: {9fed14d4-acab-6a27-6d89-070000000000}
LogonId: 0x78960
TerminalSessionId: 1
IntegrityLevel: Medium
Hashes: SHA256-D7757AF4E0F190836839ED6780001764412102E021B5939A60671A464512E9E2
ParentProcessGuid: {9fed14d4-b5b4-6a27-6202-000000001900}
ParentProcessId: 10852
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
ParentUser: IMAN999\Iman SOC
TimeCreated : 6/8/2026 11:33:16 PM
ProviderName : Microsoft-Windows-Sysmon
Id : 1
Message : Process Create:
RuleName:
UtcTime: 2026-06-09 06:33:16.879
ProcessGuid: {9fed14d4-b3ac-6a27-3e02-000000001900}
ProcessId: 4032
Image: C:\Windows\System32\cmd.exe
FileVersion: 10.0.19041.4395 (winbuild.190401.0000)
Description: Windows Command Processor
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: Cmd.Exe
CommandLine: C:\Windows\System32\cmd.exe /c "net session"
CurrentDirectory: C:\NulltackLab\
User: IMAN999\Iman SOC
LogonGuid: {9fed14d4-acab-6a27-6d89-070000000000}
LogonId: 0x78960
TerminalSessionId: 1
IntegrityLevel: Medium
Hashes: SHA256-8D0F472413C0C8C0C03F095810CA167F8C0C638597CC085F4311180E088B0
ParentProcessGuid: {9fed14d4-b38c-6a27-3302-000000001900}
ParentProcessId: 3252
ParentImage: C:\Users\Iman SOC\AppData\Local\Programs\Python\Python314\python.exe
ParentCommandLine: "C:\Users\Iman SOC\AppData\Local\Programs\Python\Python314\python.exe" NulltackKatz_v1.2.py
ParentUser: IMAN999\Iman SOC

```

Question	Answer
What was the parent process of NulltackKatz?	powershell.exe
What command-line arguments were passed?	C:\Users\Iman SOC\AppData\Local\Python\pythoncore-3.14-64\python.exe" .\NulltackKatz_v1.2.py
What user context did it run under?	IMAN999\Iman SOC

Task 5 – Network Traffic Analysis (T1041 Detection)

Open the Wireshark capture and apply the following filter:

tls and ip.dst

Question	Answer
What destination IP range received the exfiltration?	142.250.102.108 and 142.250.102.0/24
What destination port was used?	587
Is the traffic encrypted?	Yes by using protocol TLSv1.3
How could a SOC detect this without decrypting?	Monitoring the release of an unusual amount of data to mail servers within a short timeframe

Task 6 — Memory Forensics (Volatility 3)

- Run Mimikatz and keep the window open so that it leaves its mark in the random access memory (RAM).

```

mimikatz 2.2.0 x64 (oe.eo)
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd "C:\Users\iman SOC\Downloads\mimikatz-master\x64"

C:\Users\iman SOC\Downloads\mimikatz-master\x64>mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #18362 Feb 29 2020 11:13:36
.## ^ ##.  "A la Vie, A l'Amour" - (oe.eo)
## / \ ##  /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > http://blog.gentilkiwi.com/mimikatz
'## v ##'   Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'   > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz #

```

Using DumpIt to capture RAM , run the DumpIt as an Administration:

```

Windows 10 x64
C:\Users\iman SOC\Downloads\DumpIt.exe

DumpIt 3.6.20230117 (X64) (Jan 17 2023)
Copyright (C) 2007 - 2021, Matt Suiche (msuiche)
Copyright (C) 2016 - 2021, Comae Technologies DMCC <https://www.comae.com>
Copyright (C) 2022, Magnet Forensics, Inc. <https://www.magnetforensics.com/>
All rights reserved.

Thanks for using DumpIt! Always use Microsoft crash dumps!

Destination path:      \\?\C:\Users\iman SOC\Downloads\IMAN999-20260611-065432.dmp
Computer name:         IMAN999

--> Proceed with the acquisition ? [y/n] y

[+] Information:
Dump Type:            Microsoft Crash Dump

[+] Machine Information:
Windows version:      10.0.19045
MachineId:             80C14056-D274-253C-0A6F-F433370CEA33
TimeStamp:            134256344755429195
Cr3:                  0x1ad002
KdCopyDataBlock:      0xffffffff8076b310688
KdDebuggerData:       0xffffffff8076ba00b20
KdpDataBlockEncoded:  0xffffffff8076ba50d80

Current date/time:     [2026-06-11 (YYYY-MM-DD) 6:54:35 (UTC)]
+ Processing... Done.

Acquisition finished at: [2026-06-11 (YYYY-MM-DD) 6:54:56 (UTC)]
Time elapsed:          0:21 minutes:seconds (21 secs)

Created file size:     4293980160 bytes (4095 Mb)
Total physical memory size: 4095 Mb

NtStatus (troubleshooting): 0x00000000
Total of written pages: 1048333
Total of inaccessible pages: 0
Total of accessible pages: 1048333

SHA-256: D98CF31EAFD19221F27E6368F6FE68344325E2CFCAA8692DFE2D36E9F7A89D4F

JSON path:             C:\Users\iman SOC\Downloads\IMAN999-20260611-065432.json

```

- Identify image info :
 - python vol.py -f APT.dmp windows.info

```

C:\Windows\System32\cmd.exe - powershell
PS C:\Users\imam SOC\Downloads\volatility3-develop> python vol.py -f APT.dmp windows.info
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
Variable Value

Kernel Base 0xf8076ae0000
DTB 0xiad000
Symbols file:///C:/Users/imam SOC/Downloads/volatility3-develop/volatility3/symbols/windows/ntkrnlmp.pdb/F57E740B088E5056E8AF0772F1CC5BEB-1.json.xz
Is64Bit True
IsPAE False
Layer_name 0 WindowsIntel32e
Memory_layer 1 WindowsCrashDump64Layer
Base_layer 2 FileLayer
KdVersionBlock 0xf8076ba0f400
Major/Minor 15.19041
MachineType 34404
KeNumberProcessors 2
SystemTime 2026-06-11 06:54:35+00:00
VtSystemRoot C:\Windows
VtProductType NtProductWinNt
VtMajorVersion 10
VtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sat Feb 2 23:04:03 1985
PS C:\Users\imam SOC\Downloads\volatility3-develop>

```

- List processes:
 - python vol.py -f APT.dmp windows.pslist

```

C:\Windows\System32\cmd.exe - powershell
PS C:\Users\imam SOC\Downloads\volatility3-develop> python vol.py -f APT.dmp windows.pslist
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
er 4 0 System 0xb7022f081040 123 - N/A False 2026-06-11 06:22:41.000000 UTC N/A Disabled
54 92 4 Registry 0xb7022f0a5080 4 - N/A False 2026-06-11 06:22:35.000000 UTC N/A Disabled
328 4 smss.exe 0xb7022fee0040 2 - N/A False 2026-06-11 06:22:41.000000 UTC N/A Disabled
436 424 csrss.exe 0xb702306ea140 11 - 0 False 2026-06-11 06:22:43.000000 UTC N/A Disabled
524 424 wininit.exe 0xb70230fce140 1 - 0 False 2026-06-11 06:22:43.000000 UTC N/A Disabled
532 516 csrss.exe 0xb70230fd0080 12 - 1 False 2026-06-11 06:22:43.000000 UTC N/A Disabled
596 516 winlogon.exe 0xb7023360f080 6 - 1 False 2026-06-11 06:22:43.000000 UTC N/A Disabled
664 524 services.exe 0xb70230f2e080 6 - 0 False 2026-06-11 06:22:43.000000 UTC N/A Disabled
676 524 lsass.exe 0xb70230fcf080 10 - 0 False 2026-06-11 06:22:43.000000 UTC N/A Disabled
788 664 svchost.exe 0xb702336c0200 19 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
812 596 fontdrvhost.exe 0xb702336c3080 5 - 1 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
820 524 fontdrvhost.exe 0xb702336c2140 5 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
912 664 svchost.exe 0xb70233765200 14 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
960 664 svchost.exe 0xb702337ab240 5 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
388 596 dwm.exe 0xb70233809240 15 - 1 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
460 664 svchost.exe 0xb70233815240 2 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
516 664 svchost.exe 0xb70233826300 3 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
724 664 svchost.exe 0xb70233820080 4 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1124 664 svchost.exe 0xb7023389a240 6 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1140 664 svchost.exe 0xb7023389c280 2 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1148 664 svchost.exe 0xb7023389e300 8 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1156 664 svchost.exe 0xb7023389d080 6 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1312 664 svchost.exe 0xb7023396b240 7 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1324 664 svchost.exe 0xb7023396b2c0 4 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1364 664 svchost.exe 0xb7023396e2c0 2 - 0 False 2026-06-11 06:22:44.000000 UTC N/A Disabled
1392 664 svchost.exe 0xb70233a052c0 4 - 0 False 2026-06-11 06:22:45.000000 UTC N/A Disabled

```

- Find Mimikatz in memory:
 - python vol.py -f APT.dmp windows.psscan | Select-String "mimikatz"

```

PS C:\Windows\system32> cd "C:\Users\imam SOC\Downloads\volatility3-develop"
PS C:\Users\imam SOC\Downloads\volatility3-develop> python vol.py -f APT.dmp windows.psscan | Select-String "mimikatz"
Progress: 100.00 PDB scanning finished
10288 7376 mimikatz.exe 0xaa0dea574080 2 - 1 False 2026-06-11 18:59:17.000000 UTC N/A Disabled

```

- Find NulltackKatz in memory:
 - python vol.py -f APT.dmp windows.psscan | Select-String "NulltackKatz"

```
PS C:\Users\iman SOC\Downloads\volatility3-develop> python vol.py -f APT.dmp windows.psscan | Select-String "NulltackKatz"
PS C:\Users\iman SOC\Downloads\volatility3-develop> python vol.py -f APT.dmp windows.psscan | Select-String "NulltackKatz_v1.2"
PS C:\Users\iman SOC\Downloads\volatility3-develop> d
```

- Check LSASS handle access:
 - python vol.py -f APT.dmp windows.handles | Select-String "lsass"

```
PS C:\Users\iman SOC\Downloads\volatility3-develop> python vol.py -f APT.dmp windows.handles | Select-String "lsass"
Progress: 100.00 PDB scanning finished
4 System 0xb70230fcf080 0x880 Process 0x1fffff lsass.exe Pid 676
4 System 0xb70230fcf080 0x884 Process 0x102a lsass.exe Pid 676
4 System 0xb70230fcf080 0x888 Process 0x102a lsass.exe Pid 676
4 System 0xb70230fcf080 0x88c Process 0x1028 lsass.exe Pid 676
4 System 0xb70230fcf080 0x894 Process 0x102a lsass.exe Pid 676
4 System 0xb70230fcf080 0x8a8 Process 0x1478 lsass.exe Pid 676
4 System 0xb70230fcf080 0x8ac Process 0x102a lsass.exe Pid 676
436 csrss.exe 0xb70230fcf080 0x284 Process 0x1fffff lsass.exe Pid 676
524 wininit.exe 0xb70230fcf080 0x190 Process 0x1fffff lsass.exe Pid 676
676 lsass.exe 0xb70230fe4160 0x4 Event 0x1f0003 -
676 lsass.exe 0xb70230fe4260 0x8 Event 0x1f0003 -
676 lsass.exe 0xb7023060c790 0xc WaitCompletionPacket 0x1 -
676 lsass.exe 0xb70233664500 0x10 IoCompletion 0x1f0003 -
676 lsass.exe 0xb702303ac060 0x14 TpworkeFactory 0xf00ff -
676 lsass.exe 0xb7023060bea0 0x1c WaitCompletionPacket 0x1 -
676 lsass.exe 0xb7022feb32c0 0x20 IRTimer 0x10002 -
676 lsass.exe 0xb7023060ca00 0x24 WaitCompletionPacket 0x1 -
676 lsass.exe 0xb70233663280 0x28 EtwRegistration 0x804 -
676 lsass.exe 0xb70233663a60 0x2c EtwRegistration 0x804 -
676 lsass.exe 0xb70233663c20 0x30 EtwRegistration 0x804 -
676 lsass.exe 0xc08c9ccc740 0x34 Directory 0x3 KnownDlls
676 lsass.exe 0xb70230fe4d60 0x38 Event 0x1f0003 -
676 lsass.exe 0xb70230fe43e0 0x3c Event 0x1f0003 -
676 lsass.exe 0xb7023365f710 0x40 File 0x100020 \Device\HarddiskVolume3\Windows\System32
676 lsass.exe 0xb70233664880 0x44 EtwRegistration 0x804 -
676 lsass.exe 0xb702301c7a90 0x48 Mutant 0x1f0001 SM0:676:304:WlStaging_02
676 lsass.exe 0xb70233613ac0 0x4c ALPC Port 0x1f0001 -
676 lsass.exe 0xc08c9857380 0x50 Directory 0xf BaseNamedObjects
676 lsass.exe 0xb7022f5d33d0 0x54 Semaphore 0x1f0003 SM0:676:304:WlStaging_02_p0
676 lsass.exe 0xb7022f5d30b0 0x58 Semaphore 0x1f0003 SM0:676:304:WlStaging_02_p0h
676 lsass.exe 0xb702307558e0 0x5c Event 0x1f0003 -
676 lsass.exe 0xb7023060cee0 0x60 WaitCompletionPacket 0x1 -
676 lsass.exe 0xb702336637c0 0x64 EtwRegistration 0x804 -
676 lsass.exe 0xb70233664160 0x68 EtwRegistration 0x804 -
676 lsass.exe 0xb70233663520 0x6c EtwRegistration 0x804 -
676 lsass.exe 0xb70233664340 0x70 IoCompletion 0x1f0003 -
676 lsass.exe 0xb702303abb0 0x74 TpworkeFactory 0xf00ff -
676 lsass.exe 0xb7022feb34e0 0x78 IRTimer 0x10002 -
676 lsass.exe 0xb7023060bdd0 0x7c WaitCompletionPacket 0x1 -
676 lsass.exe 0xb7022feb3920 0x80 IRTimer 0x10002 -
676 lsass.exe 0xb7023060d220 0x84 WaitCompletionPacket 0x1 -
676 lsass.exe 0xb7022f5d3d10 0x88 Mutant 0x1f0001 -
676 lsass.exe 0xb70230755960 0x8c Semaphore 0x100003 -
676 lsass.exe 0xb70230755160 0x90 Semaphore 0x100003 -
676 lsass.exe 0xb70233639aa0 0x94 ALPC Port 0x1f0001 SelsaCommandPort
676 lsass.exe 0xb70233602ac0 0x98 ALPC Port 0x1f0001 -
676 lsass.exe 0xb70230f43df0 0x9c ALPC Port 0x1f0001 -
676 lsass.exe 0xc08c0c0cb60 0xa0 Key 0xf003f MACHINE
676 lsass.exe 0xb7023365f40 0xa4 Event 0x1f0003 -
```

Question	Answer
What is the PID of Mimikatz in memory?	10288
Does the memory dump show the executed commands?	yes

MITRE ATT&CK Mapping

Technique ID	Technique Name	Phase	Detection Method
T1059.006	Command and Scripting Interpreter:Python	Execution	When Python executed (NulltackKatz_v1.2.py), it was detected by Sysmon event ID 1.
T1003.001	OS Credential Dumping:LSASS Memory	Credential Access	Memory Forensics and Handles Audit: Volatility 3 discovered an abnormal handle with access mask (0x1fff) that was targeting lsass.exe PID 676
T1082	System Information Discovery	Discovery	Host-Based Audit Logs: Sysmon recorded internal discovery commands executed by the script such as net user,ipconfig /all, and tasklist
T1071.001	Command and Control	Application Layer Protocol: Web Protocols	Network Traffic Analysis: Captured via packet analysis and pcap displaying active outbound connections to the external malicious IP addresses.

Detection Rule Writing

Suricata Rule — DNS Exfiltration to Gmail

```
alert dns any any -> any any (  
  msg:"DNS Exfiltration targeting Gmail";  
  dns.query;  
  content:"gmail.com";  
  nocase;  
  sid:1000001;  
  rev:1
```

)

Sysmon Detection Rule Checklist

A complete detection strategy should include the following Sysmon event IDs:

Event ID	Event Name	What to Detect / Rule Logic
Event ID 1	Process Creation	Monitoring python.exe tool runs, especially when it calls unknown scripts in the downloads folders, such as NulltackKatz_v1.2.py. I've also configured a rule to detect any direct execution of a file named mimikatz.exe or its digital signatures (hash/signatures)
Event ID 10	Process Access	Monitor any process that tries to establish a direct connection to read the sensitive system memory of lsass.exe. flag suspicious requests for full access privileges such as (the masks 0x1ffff or 0x1010)that are commonly used to steal hashes and passwords.
Event ID 3	Network Connection	Monitor non-network-based processes like Python scripts or local programs, especially when they suddenly try to open encrypted external connections to email servers and smuggle ports like 587 or 465 (SMTP over TLS) to leak data.
Event ID 7	Image Loaded	The Python interpreter or any suspicious process suddenly downloading system-sensitive software libraries (DLLs), especially libraries used to access the kernel or encrypt and transmit data over the network to synchronize the operation of the malicious tool.

Task 7 APT28 Full Threat Intelligence Study

Research Areas Part 1:

Group Identity & Attribution

- What are all known aliases for APT28 (Fancy Bear, BlueDelta, etc.)?
 - APT28 carries a range of names, each one shaped by the cybersecurity vendor or intelligence organization tracking the group.

Alias	Organization
Fancy Bear	CrowdStrike
Sofacy / Sofacy Group	Kaspersky
Sednit	ESET
Pawn Storm	Trend Micro
STRONTIUM / Forest Blizzard	Microsoft
BlueDelta	Recorded Future
IRON TWILIGHT	SecureWorks
FROZENLAKE	Google
Tsar Team	iSight Partners
Snakemackerel	iDefense
Swallowtail	Symantec
ITG05	IBM
Fighting Ursa	Palo Alto Networks
G0007	MITRE ATT&CK
UAC-0028	CERT-UA

- What is the exact military unit number and chain of command?
 - Many people believe that GRU Unit 26165, a cyber espionage unit in the Main Directorate of the General Staff of the Armed Forces of the Russian Federation (GRU), is responsible for APT28. Cyber intelligence and credential theft activities against government, military, political, and strategic targets are the responsibilities of this unit.

- What is the primary motivation and strategic objectives?

Strategic Objectives

- Conduct intelligence gathering against foreign governments and institutions.
- Steal credentials and sensitive information from targeted organizations.
- Monitor political, diplomatic, and military activities.
- Support Russian national security and foreign policy interests.
- Get long-term access to victim networks for information gathering and surveillance.
- When in line with strategic goals, carry out information and influence activities.

- What other GRU units operate alongside APT28?

GRU Unit	Common Name	Primary Role
Unit-26165	APT28 / Fancy Bear	Theft of credentials, cyber espionage, and intelligence collection
Unit-74455	Sandworm Team	Destructive and disruptive cyber activities
Unit-54777	Information Operations Unit	Psychological operations and information warfare

- What is the relationship with Sandworm Team (GRU Unit 74455)?

APT28	Sandworm Team
GRU-Uni- 26165	GRU-Unit-74455
Cyber espionage	Cyber disruption and sabotage
Credential theft	Destructive malware operations
Intelligence gathering	Critical infrastructure attacks
Long-term covert access	Operational impact and disruption

Part 2: Historical Operations

Operation	Year	Target	Techniques Used
U.S. Presidential Election Interference	2016	DNC, Hillary Clinton campaign	T1566.002 -(Spear-phishing Link.) T1003.001 -(LSASS Memory Dumping.) T1041 -(Exfiltration Over C2 Channel.)
WADA / USADA Operations	-2014 2018	Anti-doping agencies	T1586.002 –Compromised Cloud Accounts.) T1566.001 –(Spear-phishing Attachment.) T1071.001-(Web Protocols.)
OPCW / Spiez Laboratory Attacks	2018	Chemical weapons investigators	T1499.004 –(Endpoint Denial of Service.) T1566.002 –(Spear-phishing Link.) T1059.006 –(Python Execution.)
BlueDelta Credential Harvesting	2025	Turkish energy, European think tanks	T1110.001-(Brute Force: Password Guessing.) T1003.001 –(OS Credential Dumping.) T1071.001-(HTTP/S C2 Communication.)
One additional operation of your choice	2015	German Federal Parliament infrastructure	T1059.003-(Windows Command Shell.) T1574.002-(DLL Side-Loading.) T1020-(Automated Data Exfiltration.)

Part 3: Technical TTP Analysis

For each MITRE ATT&CK technique below, document how APT28 specifically uses it:

MITRE Technique ID	Technique Name	APT28 Specific Implementation
T1566.001	Spearphishing Attachment	Send spearphishing emails with malicious attachments to targets, such as weaponized PDFs or Word documents with macro capabilities (e.g., by utilizing flaws like CVE-2023-38831 to drop secondary payloads).

T1204.001	Malicious Link	Create false login portals or phishing URLs that fool victims into clicking, resulting in drive-by downloads or credential harvesting (OAuth tokens, corporate logins).
T1003.001	LSASS Memory Dumping	Harvest NTLM hashes and plaintext passwords from the Local Security Authority Subsystem Service (lsass.exe) process area via programmatic scraping, APIs, or open handles (access mask 0x1 ffff).
T1134	Token Manipulation	Duplicate or steal the current access tokens of privileged system accounts in order to raise privileges, circumvent access limitations, and perform actions as administrative users without knowing their passwords.
T1041	Exfiltration over C2	Before being carefully exfiltrated back to remote enemy infrastructure via HTTP/S, DNS, or encrypted SMTP routes, passwords, host discovery data, and internal emails should all be compressed and encrypted.
T1070.004	Redirect Victim to Legitimate Site	After successfully obtaining credentials via a phony or landing page, automatically switch the victim's browser to the actual, authentic website (such as the official webmail or portal) to hide the attack and lower suspicion.

Part 4: Infrastructure & IoCs

Research Item

- List 5 domains used by APT28 (with purpose)

Domain	Purpose
outlook.live.com	User account targeting
outlook.office.com	Email credential harvesting
outlook.office365.com	Credential interception
autodiscover-s.outlook.com	Email service redirection
imap-mail.outlook.com	Mail authentication interception

- List 5 IP addresses attributed to APT28

IP Address	Purpose
5.226.137.151	Malicious DNS / AitM Infrastructure
5.226.137.230	APT28 DNS Infrastructure
23.106.120.119	Command & Control Infrastructure
37.221.64.77	Malicious DNS Server
64.120.31.96	AitM / Credential Harvesting Infrastructure

- List 5 file hashes (MD5 or SHA256) from APT28 malware

Malware / Campaign	SHA256 Hash
Zebrocy	fae335a465bb9faac24c58304a199f3bf9bb1b0bd07b05b18e2be6b9e90d72e6
Zebrocy	eb81c1be62f23ac7700c70d866e84f5bc354f88e6f7d84fd65374f84e252e76b
Zebrocy	6e89e098816f3d353b155ab0f3377fe3eb3951f45f8c34c4a48c5b61cd8425aa
Zebrocy	aac3b1221366cf7e4421bdd555d0bc33d4b92d6f65fa58c1bb4d8474db883fec
Zebrocy	b45dc885949d29cba06595305923a0ed8969774dae995f0ce5b947b5ab5fe185

Part 5: Defensive Recommendations

- Based on your research, provide:

Recommendation:

- Three specific detection rules to identify APT28 activity
 - Detect Masqueraded rundll32.exe Payload Execution: Monitor process creation events for rundll32.exe spawning from unexpected parent processes or running with suspicious command lines (loading from .tmp or hidden folders).
 - Identify Misuse of Cloud Storage and AI Platforms for C2: Look for endpoint-initiated outbound network connections or DNS requests targeting domains of non-corporate cloud file-sharing services (like Koofr/Icedrive) or API mocking platforms.
 - Identify Hijacked Router and Malicious DNS Traffic: Monitor DNS logs and NetFlow data for high volumes of DNS queries originating from internal routers going to unknown or suspicious external attacker-controlled VPSs.

- Three architectural changes to prevent APT28-style attacks
 - Zero Trust Architecture (ZTNA): Use network segmentation and ongoing user verification to put Zero Trust principles into practice. After a first compromise, this restricts lateral movement and keeps attackers from gaining access to vital systems.
 - Secure Edge Devices: Limit administration access, apply frequent updates, and disable direct internet exposure to safeguard routers, firewalls, and remote access devices. Infrastructure compromise is less likely as a result.
 - program Control and Host Hardening: Use program whitelisting tools such as AppLocker or Windows Defender Application Control to prevent unauthorized software execution. This prevents harmful scripts and malware from operating on endpoints.

References & Resources

MITRE ATT&CK	https://attack.mitre.org
Suricata Docs	https://docs.suricata.io/en/latest/rules/
Volatility 3 Docs	https://volatility3.readthedocs.io
Sysmon Sysinternals	https://docs.microsoft.com/en-us/sysinternals/downloads/sysmon
Mimikatz GitHub	https://github.com/AnmaR3B/mimikatz
APT28 Group Profile	https://attack.mitre.org/groups/G0007/